

Assignment – 1.4

Q - 1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

GPU Brand	NVIDIA
GPU Model Numbers	GeForce RTX 3060
GPU Type	Dedicated(Discrete) GPU

Q - 2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



Component	Function
CPU Fan	Blows air to cool the CPU.
Heatsink	Absorbs and removes heat from the CPU.
Case Fan	Pushes hot air out and brings cool air into the PC.
GPU Fan	Cools the graphics card during gaming or heavy work.

Q – 3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Feature	Air Cooling	Liquid Cooling
Cost	Lower	Higher
Cooling Efficiency	Good	Excellent
Maintenance	Very Low	Higher (pump/coolant checks)
Noise	Can be louder under load	Usually quieter
Installation	Easy	More complex

Q – 4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

- I would choose liquid cooling for my gaming PC.
- It keeps the GPU cooler during long gaming and while streaming IPL matches at the same time.
- Better cooling helps the GPU give stable performance and reduces the chance of overheating.
- Although liquid cooling is more expensive and needs more maintenance than air cooling, it is the better choice for high-performance gaming and streaming.